Q4: A Green-Potential Refinement of Determinant Rigidity

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Abstract

For a closed negative Einstein four-manifold, let Ψ solve $(\Delta + 2a)\Psi = \det(W^+ - a\,\mathrm{Id}\,/3)$, where $\mathrm{Ric} = -ag$. An almost-Kähler structure is parallel if Ψ is nonpositive at just one of its torsion zeros. In particular, the metric-only condition $\Psi \le 0$ everywhere is sufficient. This recovers pointwise determinant rigidity and replaces its sign assumption by a global Green-potential condition. An exact identity quantifies the positive potential required at every torsion zero of any hypothetical non-Kähler closed solution. The potential condition is not proved automatic under the general negative Goldberg hypotheses.

1 The metric potential and the result

Let (M^4, g, J, Ω) be smooth, closed, connected and almost-Kähler, with the orientation induced by J, and assume

$$\mathrm{Ric} = -ag, \qquad a > 0, \qquad s = -4a, \qquad |\Omega|^2 = 2.$$

Curvature acts on two-forms; all operator norms are Hilbert–Schmidt norms. Set

$$\mathcal{R}_+ = W^+ - \frac{a}{3}\,\mathrm{Id}, \qquad A = -\mathcal{R}_+, \qquad \mathrm{tr}\,A = a, \qquad u = \sigma_2(A) = \frac{a^2}{3} - \frac{1}{2}|W^+|^2.$$

Use the nonnegative Laplacian $\Delta = -\operatorname{div}\nabla$. The strictly positive operator $L_a = \Delta + 2a$ has an inverse $\mathcal{G}_a$. Define two globally smooth scalar functions

$$\Psi = \mathcal{G}_a(\det \mathcal{R}_+), \qquad \mathcal{D} = \mathcal{G}_a(|\nabla A|^2). \tag{1}$$

These are functions determined by the metric and orientation, independently of the choice of a compatible almost-Kähler form.

Theorem 1.1. *If $q(p) = |\nabla\Omega|^2(p) = 0$ and $\Psi(p) \le 0$ at some $p \in M$, then*

$$\nabla\Omega = 0, \qquad \mathcal{R}_+ = -a\,n \otimes n, \qquad n = \Omega/\sqrt{2}.$$

Consequently, the metric-only condition $\Psi \le 0$ on all of M forces every compatible almost-Kähler form in this orientation to be parallel.

The global sign corollary uses Armstrong's torsion-zero theorem, recalled below. No condition on the Betti numbers, injectivity radius or a family of metrics is introduced. The conclusion is conditional on the displayed potential sign, not a resolution of Q4 without additional assumptions.

2 Two identities and positivity of the inverse

The scalar identity proved in the preceding determinant note [3] is

$$(\Delta + 2a)u = |\nabla A|^2 - 18 \det \mathcal{R}_+. \tag{2}$$

For completeness, the classical Einstein Weyl–Weitzenböck equation [1] is

$$\nabla^*\nabla W^+ + \frac{s}{2}W^+ - 6(W^+)^2 + 2|W^+|^2 \,\mathrm{Id} = 0.$$

Pairing with W^+ gives

$$\Delta|W^+|^2 = -s|W^+|^2 + 36 \det W^+ - 2|\nabla W^+|^2.$$

The algebraic shift yields

$$|W^+|^2 = \frac{2a^2}{3} - 2u, \qquad \det W^+ = -\frac{2a^3}{27} + \frac{a}{3}u - \det A.$$

Since $\det A = -\det \mathcal{R}_+$, these imply (2) and hence

$$\boxed{u = \mathcal{D} - 18\Psi.} \tag{3}$$

The inverse is positivity preserving. In heat-kernel notation,

$$\mathcal{G}_a f(x) = \int_0^\infty e^{-2at}(e^{-t\Delta}f)(x)\, dt = \int_M G_a(x,y) f(y)\, dV_g(y). \tag{4}$$

The heat kernel on a connected closed manifold is strictly positive for $t > 0$, and $G_a(x,y) > 0$ away from its integrable diagonal singularity. Equivalently, positivity follows from the weak and strong maximum principles for L_a. Thus

$$\mathcal{D} \geq 0, \qquad \mathcal{D}(p) = 0 \implies \nabla A \equiv 0.$$

The constant mode gives $\mathcal{G}_a 1 = 1/(2a)$; no projection onto mean-zero functions is involved.

Now suppose $q(p) = 0$. The harmonic-form equation gives

$$\nabla^*\nabla n = 2(a\,\mathrm{Id} - A)n, \qquad q = 4(a - \langle An, n\rangle). \tag{5}$$

Relative to $\mathbb{R}n \oplus n^\perp$, at p write

$$A(p) = \begin{pmatrix} a & z^* \\ z & C \end{pmatrix}, \qquad \operatorname{tr} C = 0.$$

Therefore, without any determinant assumption,

$$\boxed{u(p) = -|z|^2 - \frac{1}{2}|C|^2 \leq 0.} \tag{6}$$

In the inherited almost-Hermitian notation,

$$W^+ = \begin{pmatrix} t & v^* \\ v & -t\,\mathrm{Id}/2 + V_\Omega \end{pmatrix}, \quad t = \frac{s}{6} + \frac{q}{4}, \quad v = \eta/\sqrt{2}, \quad \operatorname{tr} V_\Omega = 0,$$

so $z = -v$, $C = -V_\Omega$ at a torsion zero. Equation (6) is exactly

$$u(p) = -\frac{1}{2}(|\eta(p)|^2 + |V_\Omega(p)|^2).$$

3 Proof of rigidity

Assume $q(p) = 0$ and $\Psi(p) \le 0$. Equations (3) and (6) show

$$0 \ge u(p) = \mathcal{D}(p) - 18\Psi(p) \ge 0.$$

Hence $u(p) = \mathcal{D}(p) = \Psi(p) = 0$ and $\nabla A \equiv 0$. The determinant of $\mathcal{R}_+$ is consequently constant. By uniqueness of L_a^{-1},

$$\Psi = \frac{\det \mathcal{R}_+}{2a}$$

is constant, and its value zero at p implies $\det \mathcal{R}_+ = 0$. Parallelness makes u constant as well, so $u = 0$. The characteristic polynomial of the symmetric operator A is therefore

$$t^3 - at^2 + ut - \det A = t^2(t - a).$$

It follows that $A^2 = aA$ and $\Pi = A/a$ is a global parallel orthogonal rank-one projection. At p, (5) gives $\Pi n = n$. Globally let $v = \Pi n$. Then

$$\nabla^*\nabla v = 2\Pi(a\,\mathrm{Id} - A)n = 0.$$

Integration gives $\nabla v = 0$, and $|v(p)| = 1$ implies $|v| = 1$ everywhere. Since $|n| = 1$ and Π is an orthogonal projection, $n = v$ throughout M. Thus $\nabla\Omega = 0$ and $A = a\,n \otimes n$.

To deduce the metric-only corollary, a torsion zero exists by Armstrong's classical observation [2, Theorem 2.2.4]. Indeed, a nowhere-zero Nijenhuis tensor, regarded as a section of $\Lambda^{0,2}T^*M \otimes T^{1,0}M$, would force

$$0 = c_2(\Lambda^{0,2}T^*M \otimes T^{1,0}M)[M] = 5\chi + 6\tau = \chi + 2(2\chi + 3\tau).$$

Both χ and $2\chi + 3\tau$ are strictly positive by the Einstein Chern–Weil formulas when $s \neq 0$. This is impossible. In the almost-Kähler class $N_J(p) = 0$ is equivalent to $q(p) = 0$. This completes the proof of Theorem 1.1.

4 The exact requirement on the remaining case

At every torsion zero, without any extra sign assumption, (3) and (6) give

$$\boxed{18\Psi(p) = \mathcal{D}(p) + \frac{1}{2}(|\eta(p)|^2 + |V_\Omega(p)|^2).} \tag{7}$$

Equivalently, with $d = \det \mathcal{R}_+$ and $d_\pm = \max\{\pm d, 0\}$,

$$\begin{aligned} 18\int_M G_a(p,y)d_+(y)\,dV_g(y) = 18\int_M G_a(p,y)d_-(y)\,dV_g(y) \\ + \int_M G_a(p,y)|\nabla A(y)|^2\,dV_g(y) \\ + \frac{1}{2}(|\eta(p)|^2 + |V_\Omega(p)|^2). \end{aligned} \tag{8}$$

Corollary 4.1. *Every hypothetical closed non-Kähler negative Einstein almost-Kähler structure satisfies $\Psi(p) > 0$ at every torsion zero p. In particular, the positive-determinant part has strictly larger $G_a(p,\cdot)$-weighted mass than the negative part at each such point.*

Proof. Equation (7) gives $\Psi(p) \ge 0$. Equality would invoke Theorem 1.1 and make the structure Kähler. Therefore the inequality is strict. □

The previous pointwise condition $\det \mathcal{R}_+ \leq 0$ implies $\Psi \leq 0$ by positivity of $\mathcal{G}_a$, so its rigidity theorem is recovered. The potential criterion does not require the determinant to be nonpositive at every point. To distinguish the two sign conditions at the scalar level, on a circle consider $f = -c + \varepsilon \cos(kx)$:

$$(\Delta + 2a)^{-1} f = -\frac{c}{2a} + \frac{\varepsilon}{k^2 + 2a} \cos(kx).$$

If $c < \varepsilon \leq c(1 + k^2/(2a))$, then f changes sign but its potential is nonpositive. This illustrates the logical relaxation for scalar functions; it does not construct an Einstein metric with that particular determinant. No additional family of Einstein metrics satisfying the relaxed condition is claimed here.

The sign of Ψ is scale invariant. Under $g \mapsto cg$, one has $a \mapsto a/c$, $\Delta \mapsto \Delta/c$, $\det \mathcal{R}_+ \mapsto c^{-3} \det \mathcal{R}_+$ and $\Psi \mapsto c^{-2}\Psi$.

Remaining Q4 statement. The argument uses no new FD or monopole estimate. It replaces the preceding pointwise sign hypothesis by a global potential condition, and identifies exactly the weighted determinant balance required by every possible non-Kähler closed solution. The unaugmented Einstein and almost-Kähler hypotheses have not been shown to force $\Psi \leq 0$ at a torsion zero; (8) permits positive determinant contribution to dominate. Thus the unrestricted closed negative Goldberg question is not settled.

References

[1] A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. **49** (1983), 405–433. `https://archive.numdam.org/item/CM_1983__49_3_405_0/`.

[2] J. Armstrong, *Almost-Kähler Geometry*, D.Phil. thesis, Oxford (1998), Theorem 2.2.4. `https://johnarmstrongmaths.com/publications/resources/thesis.pdf`.

[3] Supplied research note, *A Determinant-Sign Case of the Four-Dimensional Negative Goldberg Problem*, 13 October 2026, `Q4_Determinant_Sign_Rigidity.tex`. The scalar equation and pointwise-sign theorem are inherited inputs; the Green-potential argument and (7) are derived here.